INCH-POUND

MIL-DTL-29576B
25 June 2007
SUPERSEDING
MIL-C-29576A
2 September 1992

DETAIL SPECIFICATION

CYLINDERS, STEEL, COMPRESSED GAS, NON-SHATTERABLE, WELDED

Reactivated after 25 June 2007 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 1,800 psi (12,400 kPa) and 2,100 psi (14,500 kPa) (rated pressure) non-shatterable steel cylinders of welded construction.

1.2 Classification. Cylinders are of the following types and classes, as specified by the applicable part number (see 6.2).

1.2.1 Types.

Type A - Internal thread
Type B - External thread

1.2.2 Classes.

Class 1 - 1,800 psi (12,400 kPa) rated pressure
Class 2 - 2,100 psi (14,500 kPa) rated pressure

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 8120

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

BB-A-1034	-	Compressed Air, Breathing. (Inactive for New Design)
TT-C-490	-	Chemical Conversion Coatings and Pre-treatments for Ferrous Surfaces (Base for Organic Coatings)

COMMERCIAL ITEM DESCRIPTIONS (CIDs)

A-A-58092	-	Tape, Antiseize, Polytetrafluorethylene
A-A-59503	-	Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27210	-	Oxygen, Aviator’s Breathing, Liquid and Gas
MIL-M-43719	-	Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-101	-	Color Code for Pipelines and Compressed Gas Cylinders
MIL-STD-130	-	Identification Marking of U.S. Military Property

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MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-2219	-	Fusion Welding for Aerospace Applications
MS90389	-	Cylinder, Oxygen, Non-Shatterable, Welded, 1800 PSI (Straight, U and Spiral Types). (Inactive for New Design)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-1264	Radiographic Inspection for Soundness of Welds in Steel by Comparison to Graded ASTM E 390 Reference Radiographs

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1684AS100	-	Cylinder, Oxygen Non-Shatterable, Welded (1,800 psi)

(Copies of the above drawing are available from the Naval Air Technical Data and Engineering Service Command (NATEC), P.O. Box 357031, NASNI, Bldg 90, San Diego, CA 92135-7031 or email to: nani_governmentdrawings@navy.mil.)

PUBLICATIONS

Defense Standardization Manual

SD-6	-	Provisions Governing Qualification

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(Copies of defense standardization manuals are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS847	-	Caps and Plugs, Protective, Dust and Moisture Seal

(Copies of the above document are available online at http://www.aia-aerospace.org/pubs/ or from the Aerospace Industries Association, 1000 Wilson Blvd, Suite 1700, Arlington, VA 22209-3901.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of the above document are available online at http://www.asq.org/ or from the American Society for Quality, P.O. Box 3005, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING & MATERIALS (ASTM) INTERNATIONAL

ASTM-E8	-	Materials, Metallic, Tension Testing of
ASTM-E1742	-	Radiographic Examination

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19248-2959.)

COMPRESSED GAS ASSOCIATION (CGA)

Pamphlet C-1	-	Methods for Hydrostatic Testing of Compressed Gas Cylinders

(Copies of this document are available online at http://www.cganet.com/ or from the Compressed Gas Association, Inc, 4221 Walney Road, 5th Floor, Chantilly, VA 20151.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE-AMS-STD-1595	-	Qualification of Aircraft, Missile and Aerospace Fusion Welders
SAE-ARP1176	-	Oxygen System and Component, Cleaning and Packaging

(Copies of these documents are available online at http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The cylinders furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 First article. When specified (see 6.2), the samples shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials that are not covered by specifications, or that are not specifically described herein, shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended. Cylinders may be made from pierced billets, cupped plates, or seamless tubing with either welded pressure heads or spun bottoms. The pressure heads used with tubes from a particular heat shall also be manufactured from a single heat in order to simplify gunfire testing.

3.3.1 Steel. Cylinders shall be made of austenitic type corrosion resistant steel in accordance with any one of the chemical analyses shown in table I for essential elements.

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TABLE I. Chemical composition by percentage.

C (max)	Mn (max)	P (max)	S (max)	Si (max)	Mo (min)	Ni (min)	Cr (min)	Ti (min)
0.12	2.00	0.04	0.04	1.00	---	8.00	17.0	0.60
0.08	2.00	0.04	0.04	1.00	---	8.00	17.0	0.40
0.08	2.00	0.04	0.04	1.00	2.00	10.0	16.0	0.40

3.4 Design and construction. The design and construction of the cylinders shall be in accordance with the applicable part number of Drawing 1684AS100 or MS90389 as specified in the contract or order (see 6.2) and shall be of welded construction. The cylinders shall have been subjected to the initial permanent volumetric expansion test within 18 months prior to delivery.

3.4.1 Wire wrapping. Unless otherwise specified in the contract or applicable drawings, wire wrapping shall not be used in the construction of cylinders furnished under this specification.

3.4.2 Welding. Welding shall be performed in accordance with MIL-STD-2219 by a welder qualified in accordance with SAE-AMS-STD-1595. Welded areas shall be subjected to radiographic inspection.

3.4.3 Neck. Shrink rings or supporting collars shall not be used for strengthening the neck of the cylinder.

3.4.4 Closure. Unless otherwise specified herein, the threaded port of each cylinder shall be closed by a metal, removable, threaded plug or cap conforming to NAS847 or a commercially equal metal plug or cap, to protect the threads and prevent entrance of foreign matter or moisture.

3.5 Performance.

3.5.1 Visual examination and dimensional check.

3.5.1.1 Visual examination. When visually examined as specified in 4.6.1.1, the cylinders shall conform to the requirements of this specification. Defects are classified in table II.

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TABLE II. Classification of defects for visual examination of the cylinder.

Defects	Critical	Minor
Incorrect outlet configuration	X	
Incorrect or defective threads	X	
Incorrect, incomplete, or faulty finish		X
Incorrect part number, or cylinder code number	X	
Flaw in material	X	
Incorrect or incomplete marking		X
Incorrect color		X
Abnormally distorted or nonuniform weld	X	
Loose particles inside the cylinder	X	
Rust or scale	X	
Irregularities in the internal surface greater than 1/16 inch	X	
Existence of any foreign material, solid or liquid	X	

3.5.1.2 Dimensional check. When dimensionally checked as specified in 4.6.1.2, the cylinders shall conform to the requirements of Drawing 1684AS100 or MS90389, as applicable. Any measurement deviating from the specified dimensions and tolerances shall be classified as a dimensional defect.

3.5.2 Cleanliness.

3.5.2.1 Non-volatile residue. When tested as specified in 4.6.2.1, the residue from the solvent used for the rinse shall be not greater than that of the clean solvent by more than 0.0015 gram per square foot of part surface area.

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3.5.2.2 Particulate. When tested as specified in 4.6.2.2, the solvent rinse shall not contain more than 5 particles between 100 and 175 micrometers per 100 milliliter (ml) of solvent; and no particles larger than 175 micrometers per 100 ml of solvent.

3.5.2.3 Halogen. When tested as specified in 4.6.2.3, there shall be no evidence of cleaning compound residue.

3.5.3 Leakage. When tested as specified in 4.6.3, the cylinder shall not show any evidence of gas bubbles after 2 minutes of submergence.

3.5.4 Permanent volumetric expansion. When tested as specified in 4.6.4, the cylinder shall not have a permanent volumetric expansion greater than 10 percent of the total volumetric expansion.

3.5.5 Burst pressure. When tested as specified in 4.6.5, class 1 cylinders shall not burst at a pressure less than 4,500 psi (31,030 kPa) and class 2 cylinders shall not burst at a pressure less than 5,250 psi (36,200 kPa). Failure of the cylinders shall occur in a ductile manner.

3.5.6 Endurance. When tested as specified in 4.6.6, the cylinder shall not show any evidence of material failure.

3.5.7 Flattening. When tested as specified in 4.6.7, the cylinder shall not show any evidence of cracking when examined visually without magnification.

3.5.8 Macrostructure defects. When tested as specified in 4.6.8, the cylinder shall show no evidence of abnormal segregation, pipes, cracks, seams or abnormal change in structure from the surface to the center.

3.5.9 Radiographic inspection. When tested as specified in 4.6.9, welded sections of the cylinder shall be in accordance with MIL-HDBK-1264.

3.5.10 Fragmentation resistance.

3.5.10.1 Procedure I. When tested as specified in 4.6.10.1, the cylinder shall remain in one piece.

3.5.10.2 Procedure II. When tested as specified in 4.6.10.2, the cylinder shall be allowed to separate into two pieces. The projected area of any piece shall be not less than 2.0 square inches. The cylinder shall exhibit no evidence of shattering into small irregular pieces or splinters.

3.5.11 Vibration (spiral cylinder). When tested as specified in 4.6.11, the cylinder assembly (brackets and cylinder) shall remain in one piece, and there shall be no evidence of constructural or material failure.

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3.5.12 Physical properties. When tested as specified in 4.6.12, the samples shall not have a tensile strength greater than 200,000 psi ($1.38x10^6$ kPa), and elongation shall be not less than 6 percent.

3.6 Weight. The weight of the cylinder shall be not greater than the values specified on the applicable drawings.

3.7 Color. Unless otherwise specified on the applicable drawings or the acquisition document (see 6.2), the color of the cylinders shall be in accordance with MIL-STD-101.

3.8 Finish. The external and internal surfaces shall be cleaned in accordance with TT-C-490, Method I or IV. The internal surface of cylinders fabricated of material other than corrosion-resistant steel shall be treated in accordance with TT-C-490, Type I. The external surface of all cylinders shall be painted with a protective finish that will not crack, chip, or scale during normal service life or under extreme environmental conditions. The color of the protective finish shall be as specified in 3.7.

3.9 Identification marking. The cylinder shall be marked for identification in accordance with MIL-STD-130.

3.9.1 Standard markings. The cylinders shall be permanently and legibly marked using a pressure sensitive decal located as shown on Drawing 1684AS100 and MS90389, as applicable. The adhesive on the decal shall conform to MIL-M-43719, Type I, Class 1. The markings shall be not greater than 0.125 inch (3.18 mm) high and shall consist of the following (listed sequence mandatory):

a. Manufacturer’s Name and CAGE (Commercial and Government Entity) code.

b. Government Part Number.

c. 1,800 psi (12,400 kPa) or 2,100 psi (14,500 kPa), as applicable.

d. Serial Number.

e. Date of permanent volumetric expansion (PVE) test (month and year) with identifying symbol between the month and year.

f. A blank area for the addition of subsequent PVE test dates.

3.9.2 Additional markings. The following additional markings shall be applied by stenciling or clear pressure sensitive decal in two places diametrically opposite of each other (see applicable drawings for location):

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1,800 psi (12,400 kPa) or 2,100 psi (14,500 kPa) (as applicable)

The size of the markings and the space between each row of markings shall be not less than 0.12 times the outer diameter of the cylinder. The adhesive on the decal shall conform to MIL-M-43719, Type I, Class 1.

3.10 Workmanship. The cylinders shall be uniform in quality and shall be free from irregularities or defects which could adversely affect safety, performance, reliability or durability. The cylinders shall be free of oil, grease, fuel, water, dust, dirt or any other foreign matter.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. First article inspection (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in table III.

TABLE III. Qualification inspection.

Inspection	Requirement	Test Method
Visual examination	3.5.1.1	4.6.1.1
Dimensional check	3.5.1.2	4.6.1.2
Cleanliness (Oxygen cylinders)	3.5.2.1 3.5.2.2 3.5.2.3	4.6.2.1 4.6.2.2 4.6.2.3
Leakage	3.5.3	4.6.3
Permanent volumetric expansion	3.5.4	4.6.4

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TABLE III. Qualification inspection - Continued.

Inspection	Requirement	Test Method
Burst pressure	3.5.5	4.6.5
Endurance	3.5.6	4.6.6
Flattening	3.5.7	4.6.7
Macrostructure defects	3.5.8	4.6.8
Radiographic inspection	3.5.9	4.6.9
Fragmentation resistance	3.5.10.1 or 3.5.10.2, as applicable	4.6.10.1 or 4.6.10.2, as applicable
Vibration (spiral cylinder)	3.5.11	4.6.11
Physical properties	3.5.12	4.6.12

4.2.1 Qualification samples. Qualification samples shall consist of the following:

a. Nine cylinders of the type (part number) for which qualification testing has been authorized.

b. Four material samples in accordance with ASTM-E8, cut from 4 complete cylinders.

c. Two sets of manufacturer's drawings in accordance with SD-6.

d. When applicable, certification from the original manufacturer that the distributor requesting qualification status is authorized to distribute the item, or to rebrand and distribute the item, as applicable.

Samples shall be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-DTL-29576 and number under authorization (reference authorizing letter and number) (see 6.3).

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4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table IV, listed sequence mandatory. If there is no Qualified Products List for the part number being procured, first article testing shall consist of the tests specified in table III.

TABLE IV. First article inspection.

Inspection	Requirement	Test Method
Visual examination	3.5.1.1	4.6.1.1
Dimensional check	3.5.1.2	4.6.1.2
Cleanliness (Oxygen cylinders)	3.5.2.1 3.5.2.2 3.5.2.3	4.6.2.1 4.6.2.2 4.6.2.3
Leakage	3.5.3	4.6.3
Permanent volumetric expansion	3.5.4	4.6.4
Burst pressure	3.5.5	4.6.5
Flattening	3.5.7	4.6.7
Radiographic inspection	3.5.9	4.6.9

4.3.1 First article samples. Unless otherwise specified herein, as soon as practicable after award of a contract or order, the manufacturer shall submit four cylinders of each part number specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these cylinders from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to conformance inspection. The first article samples shall be furnished to the Government as specified in the contract (see 6.2).

4.3.1.1 First article information. Upon completion of the first article inspection, the Government activity responsible for conducting the inspection (see 6.2) shall report the results of the inspection to the contracting officer with appropriate recommendations.

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4.3.1.2 First article sample disposition. The samples will be destroyed during the inspection program and will not be returned to the contractor.

4.4 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in table V, listed sequence mandatory. The sampling and inspection levels and acceptance criteria shall conform to ANSI/ASQ-Z1.4.

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Cylinder. An inspection lot size shall be expressed in units of one cylinder of the same size, made essentially under the same conditions and from the same materials and components. The sample unit shall be one cylinder.

4.4.1.2 Sampling for tests and examinations of cylinders. The sample size, acceptance criteria, tests, and examinations required for the cylinder shall be as specified in table V.

4.5 Test conditions.

4.5.1 Gas. Unless otherwise specified herein, the gas used in testing the cylinders shall be either oxygen conforming to MIL-PRF-27210, Type I; nitrogen conforming to A-A-59503, Type I, Class 1, Grade B; or air conforming to BB-A-1034.

4.5.2 Temperature and pressure. Unless otherwise specified herein, tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. The temperature and barometric pressure shall be recorded at the time of inspection and, when required, the tests results shall be corrected to normal temperature and pressure (NTP) conditions. NTP conditions are 70 °F (21 °C) and 29.92 inches (101.3 kPa) of mercury .

4.5.3 Antiseize tape. To prevent damage to the cylinder threads or valves, male pipe threads of fittings installed in cylinders during tests shall be properly coated with antiseize tape conforming to A-A-58092 before being installed in the cylinder. The tape shall be applied according to instructions specified in A-A-58092. Tape shall be prevented from entering the inside of the fittings.

4.6 Inspection methods.

4.6.1 Visual examination and dimensional check.

4.6.1.1 Visual examination. Every cylinder shall be visually examined for conformance to the requirements of 3.5.1.1. Defects shall be as specified in table II.

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TABLE V. Conformance inspection.

Inspection	Method	Sample Size	Acceptance criteria
Visual examination	4.6.1.1	a. Every cylinder for critical defects. b. Inspection Level II for minor defects.	a. Reject all units with any critical defect. b. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensional check	4.6.1.2	1/ Inspection Level S-2	Acceptance number zero, rejection number 1.
Cleanliness	4.6.2.1 4.6.2.2 4.6.2.3	1/ Inspection Level S-3	Acceptance number zero, rejection number 1.
Leakage	4.6.3	Every cylinder	Reject all defective units.
Permanent volumetric expansion	4.6.4	Every cylinder	Reject all defective units.
Burst pressure 2/	4.6.5	1/ Inspection Level S-2	Acceptance number zero, rejection number 1.
Flattening 2/	4.6.7	1/ Inspection Level S-1	Acceptance number zero, rejection number 1.
Radiographic inspection	4.6.9	Every cylinder	Reject all defective units.
Physical properties 2/	4.6.12	1/ Inspection Level S-1	Acceptance number zero, rejection number 1.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specific inspection level of ANSI/ASQ-Z1.4.
2/ This is a destructive test (see 6.2).

4.6.1.2 Dimensional check. The cylinders shall be dimensionally checked for conformance to the requirements of 3.5.1.2.

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4.6.2 Cleanliness (oxygen cylinders). The cleanliness test shall consist of the non-volatile residue test (see 4.6.2.1), the particulate test (see 4.6.2.2), and the halogen test (see 4.6.2.3).

4.6.2.1 Non-volatile residue test.

4.6.2.1.1 Preparation of cylinder. A clean plug shall be placed in the outlet of the cylinder undergoing test. The surface around the plug and the cylinder outlet shall be cleaned with an oxygen-safe cleaning compound, then dried using a clean lint-free cloth.

4.6.2.1.2 Procedure. Non-volatile residue shall be determined in accordance with SAE-ARP1176 and the following:

a. Internal area of the cylinder shall be the basis for determining the quantity of cleaning compound required.

b. The plugged cylinder shall be agitated for not less than 10 minutes in a manner that completely wets the inner walls with the cleaning compound.

c. Particulate content shall be determined in accordance with 4.6.2.2.

d. Non-volatile residue shall be in conformance with 3.5.2.1.

4.6.2.2 Particulate test. Upon successful completion of the test specified in 4.6.2.1, examine the remaining one-half of the solvent used for the rinse for particles. To determine the particle content, filter the solvent through a 1.2 micromillipore, or equivalent membrane filter and inspect the filter surface for conformance to the requirements of 3.5.2.2. Upon successful completion of this test, the cylinders shall be subjected to the test specified in 4.6.2.3.

4.6.2.3 Halogen test. Upon successful completion of the test specified in 4.6.2.2, the cylinders shall be tested to detect any solvent residue. A halogen or halide detector, having a sensitivity of not less than 3×10^{-4} standard cubic centimeters per second, shall be used for this test. The cylinders shall conform to the requirements of 3.5.2.3.

4.6.3 Leakage. The cylinder shall be charged to rated service pressure, 1,800 +50, -0 psig (12,400 +345, -0 kPa) or 2,100 +50, -0 psig (14,500 +345, -0 kPa). With this pressure maintained, the cylinder shall be submerged in water for not less than 2 minutes. All parts of the cylinder shall be covered with not less than 1 inch (25.4 mm) of water. The cylinder shall pass the requirements specified in 3.5.3.

4.6.4 Permanent volumetric expansion. Permanent volumetric expansion of the cylinder shall be determined as specified in the hydrostatic test of Pamphlet C-1 of the Compressed Gas Association. The cylinder shall pass the requirements specified in 3.5.4, herein. The cylinder shall have passed this test within 18 months prior to delivery.

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4.6.5 Burst pressure.

4.6.5.1 Class 1 cylinders. The cylinder shall be hydrostatically pressurized at a rate not greater than 25,000 psig (172,650 kPa) per minute to a maximum pressure of 4,350 psig (29,989 kPa), and the pressure maintained for three minutes. The pressure shall then be gradually increased to destruction and the burst pressure recorded. The cylinder shall conform to the requirements of 3.5.5.1.

4.6.5.2 Class 2 cylinders. The cylinder shall be hydrostatically pressurized at a rate not greater than 25,000 psig (172,650 kPa) per minute to a maximum pressure of 5,000 psig (34,530 kPa), and the pressure maintained for three minutes. The pressure shall then be gradually increased to destruction and the burst pressure recorded. The cylinder shall conform to the requirements of 3.5.5.2.

4.6.6 Endurance test. The cylinder shall be hydrostatically charged to 1,800 +50, -0 psig (12,400 +345, -0 kPa) or 2,100 +50, -0 psig (14,500 +345, -0 kPa), as applicable, and the pressure released to obtain an internal pressure of 50 $\pm$ 50 psig (345 $\pm$ 345 kPa). This shall constitute one cycle. The charging and discharging shall be repeated for 20,000 cycles. The cylinder shall pass the requirements specified in 3.5.6. The cylinder shall then be subjected to and pass the permanent volumetric expansion test.

4.6.7 Flattening. The cylinder shall be flattened to six times the wall thickness between wedge-shaped knife edges with a 60-degree (1.05 rad) included angle and rounded to a 0.5 inch (13mm) radius. The knife edge shall be applied perpendicularly to the longitudinal welds, the bend knuckle shall occur on the weld. The cylinder shall pass the requirements specified in 3.5.7.

4.6.8 Macrostructure (deep acid etch). Longitudinal sections from the cylinder shall be etched in an aqueous solution containing 50 percent concentrated hydrochloric acid by volume and maintained at a temperature of 160 $\pm$10 °F (71 $\pm$5.5 °C) for sufficient time to show the presence of defects. The sections shall then be visually examined by a metallographist and shall meet the requirements specified in 3.5.8.

4.6.9 Radiographic inspection. The welded sections of cylinder, except brackets welded to the spiral cylinder, shall be inspected in accordance with ASTM-E1742. The welded sections shall meet the requirements specified in 3.5.9.

4.6.10 Fragmentation resistance. Oxygen conforming to MIL-PRF-27210, Type I (see 4.5.1) shall be used for this test.

4.6.10.1 Procedure I. Cylinders greater than 2.5 inches (6.35 cm) in diameter shall be subjected to gunfire under the following conditions:

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a. The cylinder shall be charged to 1,800 ± 50 psig (12,400 ± 345 kPa) or 2,100 ± 50 psig (14,500 ± 345 kPa), as applicable, at gun range ambient air temperature.

b. The cylinder may be supported but not constrained.

c. The ammunition shall be .50 caliber M-2, armor-piercing.

d. The range shall be 50 yards (45.7 m) maximum.

e. The various cylinders taken for test shall be tested, each progressively, in a different position. The point of impact shall be near the weld. The cylinder positions shall be as follows:

(1) With the longitudinal axis of the cylinder normal to the line of fire.

(2) With the longitudinal axis of the cylinder 45 degrees (0.785 rad) from normal toward the gun position.

(3) With the longitudinal axis of the cylinder parallel to the line of fire with inlet port away from the gun position.

f. All shots shall be tumbled. A suggested method for projectile tumbling is show on figure 1.

g. The tumbled projectile shall have a velocity of not less than 2,600 ft/sec (792 meters/sec) at the point of impact with the cylinder.

h. To verify that the projectile was satisfactorily tumbled, the size entry hole made by the projectile shall be not less than 0.5 inch by 1.5 inches (13 mm by 39 mm).

i. Verify that the projectile trajectory and tumble are satisfactory. This may be determined by the location and visual appearance of a hole made by the passage of the projectile through vertically suspended sheets of paper at the target area.

The cylinder shall meet the requirements specified in 3.5.10.1.

4.6.10.2 Procedure II. Cylinders 2.5 inches (6.35 cm) or less in diameter shall be subjected to gunfire under the following conditions:

a. The cylinder shall be charged to 1,800 ± 50 psig (12,400 ± 345 kPa) or 2,100 ± 50 psig (14,500 ± 345 kPa) at gun range ambient air temperature.

b. The cylinder may be supported but not constrained.

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c. The cylinder shall be positioned as indicated in 4.6.10.1e.

d. The ammunition shall be a .30 caliber, armor piercing, with velocity of 2,800 ± 100 feet per second (854 ± 30 meters/sec).

e. The range shall be approximately 20 yards (18.3 m).

f. Shots shall not be tumbled.

The cylinders shall meet the requirements specified in 3.5.10.2.

4.6.11 Vibration (spiral cylinder). The vibration test shall be conducted only in the horizontal position and only at room temperature. The cylinder shall be pressurized to 1,800 ± 50 psig (12,400 ± 345 kPa) or 2,100 ± 50 psig (14,500 ± 345 kPa), as applicable, mounted to a rigid fixture by the cylinder's mounting brackets and then subjected to the vibration test specified in MIL-STD-810, Method 514.5, Procedure 1, Table 514.5-III, Figures 514.5-8. The power spectral density (W_o) shall be 0.30 g^2/Hz. The test time shall be 3 hours. The cylinder brackets shall then be examined and meet the requirements specified in 3.5.11.

4.6.12 Physical properties. Physical property testing of material samples shall consist of tensile strength and elongation testing in accordance with ASTM-E8 for conformance to the requirements of 3.5.12.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cylinders covered by this specification are intended for use in storing compressed gas.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of the specification.

b. Government part number (see 3.4).

c. National Stock Number.

d. Applicable Government drawings, including revisions (see 3.4).

e. Quantity desired.

f. Whether first article inspection is waived (see 4.3.1).

g. The name and address of the first article inspection laboratory (see 4.3.1) and the name of the Government activity responsible for conducting the first article inspection program (see 4.3.1.1).

h. Cylinders subjected to destructive tests (see table V) should not be shipped as part of the acquisition quantity.

i. Marking requirements (see 3.9).

j. Color (see 3.7).

k. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL-29576) whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command, Code 4.6.7.4, 48110 Shaw Road, Bldg 2187, Patuxent River, MD 20670-5304.

6.3.1 Drawings. When requested, the manufacturer should submit engineering drawings and inspection reports in accordance with SD-6.

6.4 First article. When a first article inspection is required, the item(s) should be a first article sample. The contracting officer should include specified instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a

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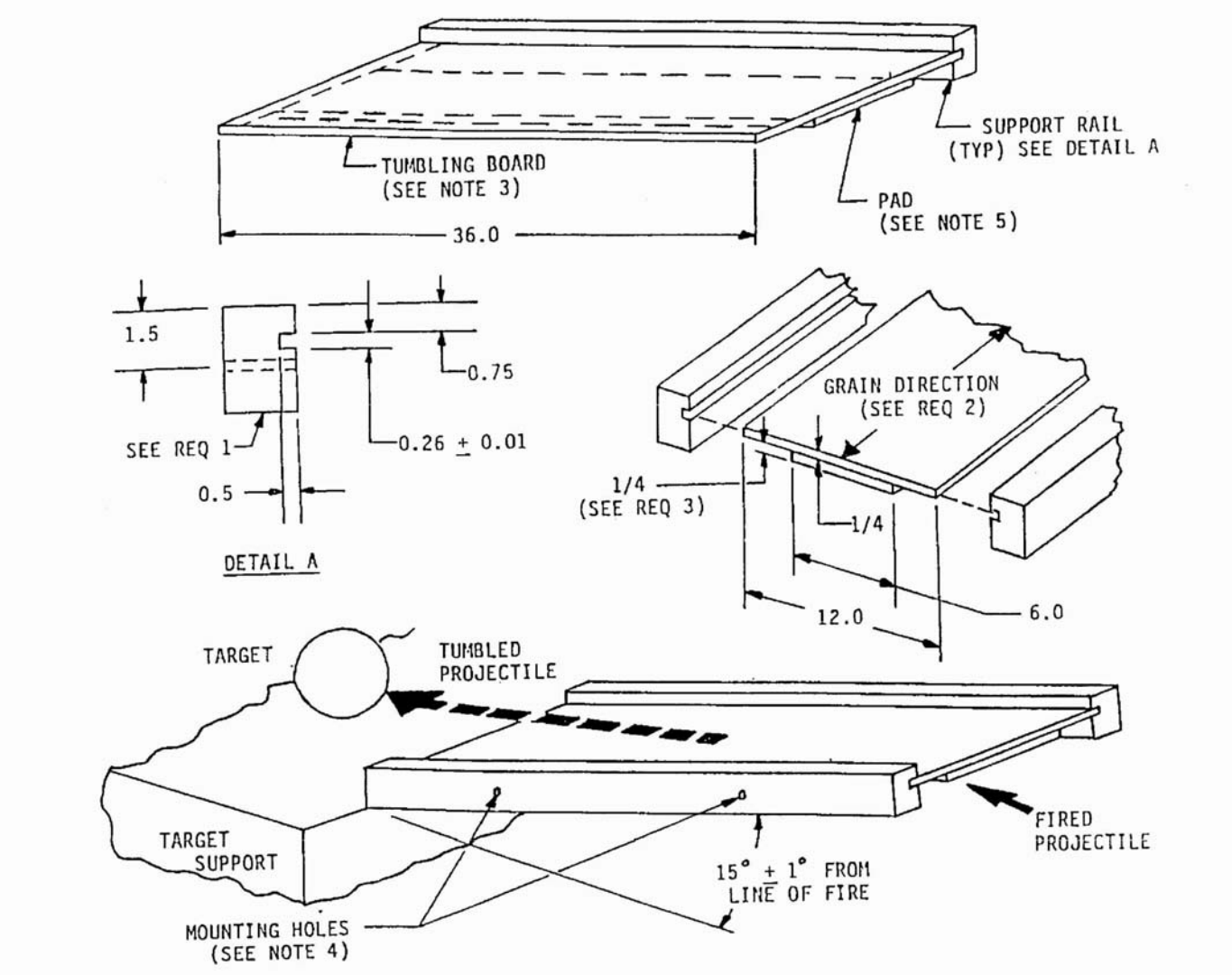
product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Color
External threads
Fragmentation resistance
Internal threads
Pressure, rated, 1,800 psi
Pressure, rated, 2,100 psi
Radiographic inspection
Trichlorotrifluorethylene
Wire wrapping

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue due to the extent of the changes.

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REQUIREMENTS:

1. Lumber, 2x3 - finish size
2. Board, plywood-grade A-A, exterior type, 3 ply
3. Sheet, rubber-styrene-butadiene (SBR), 55 ± 5 durometer (shore A)

NOTES:

1. Dimensions in inches
2. Tolerance: decimals ± 0.1, fractions ± 1/64
3. Tumbling board may be moved in the support rails for repeated firings.
4. Mounting holes method used to support tumbling board is optional, however, the board must be rigidly mounted.
5. Pad shall be cemented to the entire length of tumbling board using a commercial contact cement.

FIGURE 1. Projectile tumbling method.

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS

(Project 8120-2007-001)

Review activities:
Air Force - 11 and 71
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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